

A Reconfigurable Co-Simulation Framework for Dynamical Analysis of Grid-Connected Wave Energy Converters

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Abstract—In this work, we develop a co-simulation based software framework to evaluate the effects of grid-connected wave energy converters in power system operations that take into account system dynamics. Specifically our approach integrates the detailed hydrodynamic response of wave energy devices to equivalent electricity generation output under grid connected configurations. A Linear Power Take-Off (PTO) model is introduced, simulating the dynamic response of WECs as they connect to a modified IEEE 14-bus system. The results suggest that WECs do not significantly enhance or degrade grid stability, thus confirming their potential for seamless integration into renewable energy grids. Additionally, WEC systems offer marginal improvements in buffering sudden demand shifts, indicating its potential contribution to power grid resilience.

Index Terms—Dynamic Co-Simulation, Grid Integration, Marine Energy, Power System Dynamics, Wave Energy.

I. INTRODUCTION

Wave energy stands out as a generation source with high potential due to its consistent and predictable power supply capabilities [1], [2], especially in coastal regions and remote communities [3]. However, renewable resources are inherently intermittent and are prone to causing frequent supply-demand mismatches, thereby inducing dynamical stability issues within the grid [4]. Technologies such as wind and solar power have been extensively studied from the perspective of grid integration, leading to well-established methods and standardized approaches for analyzing their impacts upon integration with the grid [5]. However, wave energy, despite its promising potential, remains relatively underexplored. Specifically, the integration of WECs into power grids has been hampered by a lack of comprehensive models that can accurately represent both the mechanical and electrical dynamics of these systems. Although tools like *WEC-SIM* have advanced the simulation of the mechanical aspects of WECs, they do not model the electrical interactions with the grid [6].

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The absence of a standardized and easily reconfigurable tool to assess dynamical grid impacts has constrained wave-based grid impact studies to steady state, and quasi steady-state analyses that largely focus on statistical operational parameters. For example, authors in [3] investigated the reliability of power systems incorporating ocean wave energy using a Well-Being approach, showing that wave energy penetration impacts system reliability, but does not induce significant failure states. Similarly, authors in [7] conducted a case study at the Pacific Marine Energy Center (PacWave), assessing the impact of grid-connected wave farms, particularly focusing on harmonics and stability challenges. More recently, authors in [2], through steady-state topology-agnostic studies, show how wave energy can reduce balancing energy requirements and has potentially higher effective load-carrying capability. One thrust of work has analyzed the dynamical impacts and subsequent controls of grid integrated WECs [8], [9], [10] - although most of these models are application specific, need simultaneous modeling of the mechanical and the electrical side, and do not allow easy reconfiguration based on device type and converter ratings.

Our work introduces a novel framework that leverages an open-source co-simulation approach to evaluating the dynamical impacts stemming from grid integration of WECs. Specifically, we focus on the development and application of a Linear Power Take-Off (PTO) model to simulate the dynamics of a WEC as it interacts with the grid. Note that in this work, the entire mechanical side of the WECs have been modeled through pre-existing modules in *WEC-SIM* [11]. Our co-simulation framework integrates the dynamical models developed in *WEC-SIM* to the electrical components, as captured by the aforementioned linear PTO device. This framework is amenable for conducting grid-integration tests for a variety of WEC converter mechanical models. As an example, we use Reference Model 3 (RM3) for modeling the mechanical side of the WEC system and, with it, perform detailed dynamical simulations studying grid integration impacts on a modified IEEE 14-bus system. Beyond development of this portable and reconfigurable co-simulation framework, we also provide new insight into how WECs can affect grid dynamics under various

operational scenarios. The results of this study are crucial to understanding the role of WECs in future energy grids.

The remainder of this paper is organized as follows. Section II outlines the methodology for the WEC modeling, Linear PTO Model, and grid modeling software setup. Section III presents the results of the simulations under various operational scenarios. Finally, Section IV concludes the paper and discusses future research directions.

II. DESIGN OF CO-SIMULATION FRAMEWORK: PROCESS PIPELINE

In this Section, we discuss a processing pipeline of our reconfigurable co-simulation framework that leverages the unique capabilities of multiple software tools, as depicted in Figure 1. This pipeline is designed to modularize the simulation process, ensuring that each component, the WEC (mechanical side), the PTO system, and the grid, can be modeled as individual stand-alone components within its optimal environment.

A. Simulation Workflow and Software Integration

The simulation setup process begins with *WEC-SIM*, where we configure and run hydrodynamic simulations of the WEC device. The *WEC-SIM* software focuses on modeling the mechanical behavior of the WEC, specifically capturing time-series data, such as wave height and the relative position of the WEC components. These simulation runs are stored for further analysis. Once the *WEC-SIM* simulations are complete, the critical time series data are extracted: wave height, relative body position, and other relevant mechanical outputs. These data points are then fed into the custom Linear PTO model, implemented within *PowerDynamics.jl* [12]. This model simulates the interaction between the WEC and the electrical grid. The dynamic response of the PTO to wave forces is subsequently recorded and maybe used for performing studies such as those pertaining to grid stability issues. Next, with *PowerDynamics.jl* we integrate the WEC-PTO model with grid simulation parameters, including grid topology, fault events, and ramp scenarios on the grid side. This step allows for a detailed analysis of the stability of the grid under various operating scenarios, considering the presence of the grid-integrated WEC system. The pipeline ensures seamless integration between *WEC-SIM* and *PowerDynamics.jl*, facilitating efficient co-simulation and providing comprehensive information on the effects of WECs on power grid performance.

B. WEC Hydrodynamic Model

In this study, we used the *WEC-SIM* RM3 model [13], a two-body point absorber equipped with a linear PTO system. The RM3 model serves as a benchmark for wave energy research, designed to support the testing and characterization of wave energy converters. The model has been extensively validated and is widely used in the wave energy community, contributing to a standardized framework for research and development in this field such as in [3], [7]. In our work, the simulation was performed under irregular wave conditions that

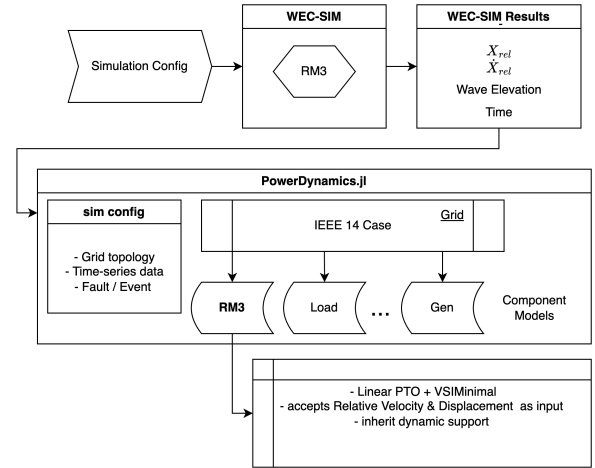


Fig. 1. Processing Pipeline of the designed Co-Simulation Framework to Study grid integration impacts of WECs.

are representative of those off the coast of Newport, Oregon, with a wave height of 3.0 meters and a wave period of 8 seconds. All relevant RM3 parameters used in this study are reported in Table I.

TABLE I
RM3 MODEL SIMULATION PARAMETERS

Parameter	Value
PTO Class	PTO1
Stiffness	0 N/m
Damping	1,200,000 Ns/m
Solver	ODE4
Step Size	0.01 seconds
Ramping	0 - 100 seconds (with and without ramp)
Wave Height	3.0 meters
Wave Period	8 seconds
Wave Spectrum	Pierson-Moskowitz (PM)
Simulation Duration	10 minutes

C. Linear PTO Model

The linear PTO model operates by simulating the PTO system as a grid-forming ac voltage source. In this context, the “VSIMinimal” model plays a critical role, as it allows precise control over the voltage amplitude and frequency. The “VSIMinimal” model is widely used in power electronics for grid integration of renewable energy sources, such as solar inverters and wind turbines. In our case, it provides a stable and reliable framework for managing the variable power output generated by the WEC as it interacts with ocean waves.

The Linear PTO model takes as input the mechanical data from the WEC simulation, specifically:

- **Relative velocity** ($\dot{X}_{rel}$): The relative velocity between the buoy and the heave plate.
- **Relative displacement** (X_{rel}): The relative displacement between the buoy and the heave plate.

Subsequently, the following key outputs are computed:

- **Mechanical power** (P_{mech}): Power generated by the PTO based on wave conditions and relative motion.

- **Electrical power** (P_{elec}): Mechanical power was converted to electrical power using an efficiency factor (η).
- **Per-unit power** (P): The electrical power normalized to per-unit (PU) values for integration with the grid model.
- **Frequency** (ω) and **Voltage** (v): The grid frequency and voltage, regulated by the droop control.

The PTO system generates a force based on the stiffness (K_{pto}) and damping (C_{pto}) properties of the PTO, as well as the relative displacement and velocity of the WEC components. Mathematically, this can be represented as:

$$F_{\text{pto}} = -K_{\text{pto}}X_{\text{rel}} - C_{\text{pto}}\dot{X}_{\text{rel}} \quad (1)$$

Subsequently, this force is used to calculate the mechanical power output of the system. Mathematically, $P_{\text{mech}} = F_{\text{pto}} \cdot \dot{X}_{\text{rel}}$. Mechanical power is then converted to electrical power with an efficiency factor η : $P_{\text{elec}} = \eta \cdot P_{\text{mech}}$, which is then normalized to generate per unit (p.u.) representations for grid integration. To maintain grid stability, the Linear PTO model uses droop control for frequency (ω_a) and voltage (V_r) regulation, dynamically adjusting output to match grid needs under varying wave conditions. Following [14], the “VSIMinimal” model represents an ac voltage source where frequency and voltage are instantaneously regulated via droop constants $K_{P,a}$ and $K_{Q,a}$, which respond to active (P) and reactive (Q) power deviations, smoothed using time constants $\tau_{P,a}$ and $\tau_{Q,a}$ for stability. The control dynamics are:

$$\dot{\phi}_a = \omega_a \quad (2)$$

$$\dot{\omega}_a = \frac{1}{\tau_{P,a}} [-\omega_a - K_{P,a} (\Re(u_a \cdot i_a^*) - P_a)] \quad (3)$$

$$\tau_{Q,a} \dot{v}_a = -v_a + V_r - K_{Q,a} (\Im(u_a \cdot i_a^*) - Q_a) \quad (4)$$

$$\dot{u}_a = \dot{v}_a e^{j\phi} + j\omega_a u_a \quad (5)$$

D. Grid Model

The electrical grid modeling for this study was conducted using the *PowerDynamics.jl* package, an open-source, dynamic grid simulation tool designed for the stability analysis of power systems with high penetration of renewable energy sources. We chose Julia for its high-performance computational capabilities and *PowerDynamics.jl* for its flexibility in integrating custom models like the linear PTO model that we leverage in this study. For our simulations, we used a modified version of the standard IEEE 14-bus test system. The IEEE 14-bus system is a well-established benchmark in power system studies, allowing consistent comparisons across different research efforts. Our modifications included the addition of a dedicated “WEC Bus” (Bus 15) connected to Bus 1 to simulate the integration of a WEC into the grid. The nameplate capacity of the added wave generator is 20 kW, contributing approximately 1.38% of the total load. Figure 2 illustrates our assumed grid topology developed from the IEEE 14-bus system.

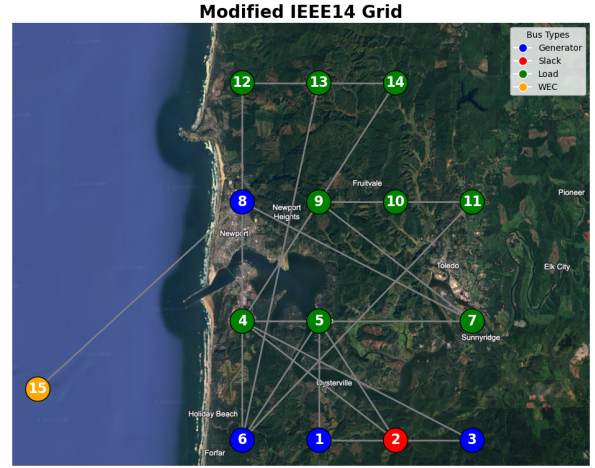


Fig. 2. Modified IEEE 14-Bus System with Integrated WEC

E. Simulation Metrics: Stability and performance

In this study, grid stability is defined by the system’s ability to maintain nominal frequency, voltage, and power levels within acceptable margins (typically within $\pm 5\%$). Stability also includes the absence of sustained oscillations or deviations that could lead to system instability. We use the following key metrics to assess grid stability and performance:

- **Frequency Deviation:** Changes in grid frequency due to variations in WEC power output.
- **Active Power Deviation:** Changes in active power production of buses.
- **Power Flow Stability:** The ability of the grid to maintain stable power flows without oscillations or instability during changes in the WEC output.
- **Frequency Stability:** The grid’s ability to maintain frequency within acceptable limits following a disturbance.
- **Voltage Stability:** The capability of the grid to maintain steady voltages at all buses after being subjected to a disturbance.
- **Transient Stability:** The grid’s ability to maintain synchronism when subjected to severe transient disturbances, such as sudden changes in WEC power output.

III. RESULTS

In this section, we present the results from a suite of experiments we conducted with our developed grid-integrated WEC model, under diverse operating conditions.

A. Flat Run Test

In this setting, the grid is assumed to operate under steady-state conditions without any disturbances. The objective is to assess the baseline behavior of the grid in both the “with WEC” and “no WEC” scenarios, focusing on active power, frequency, and power flow stability. This allows us to determine whether the presence of the WEC introduces any instability to the grid. We analyze the active power (P) and frequency (ω) at the generator buses (Buses 1, 2, 3, 6, and 8)

to establish a baseline comparison. Figures 3 and 4 depict the active power and frequency response for these buses during the flat run. Overall, the system remains stable in both scenarios, with no significant instability introduced by the WEC. The active power and frequency profiles for Buses 1, 2, 3, 6, and 8 show minimal differences between the “with WEC” and “no WEC” scenarios. The results suggest that, for a variety of scenarios, WEC integration under baseline conditions does not cause substantial deviations in grid behavior. The WEC maintains stable power flow and frequency profiles, similar to the scenario without WECs, indicating that it does not significantly improve or degrade dynamical stability, for these scenarios.

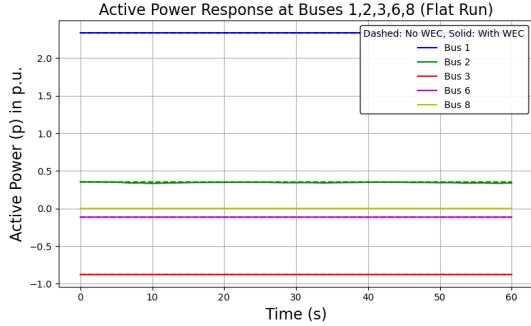


Fig. 3. Active Power at Buses 1, 2, 3, 6, and 8 during Flat Run.

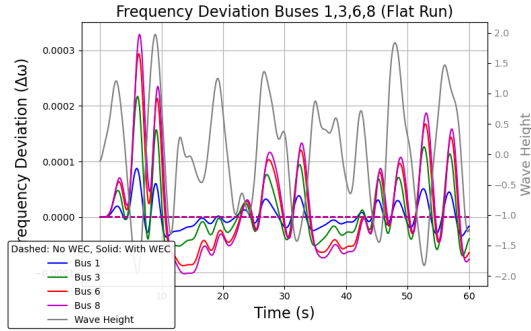


Fig. 4. Frequency at Buses 1, 3, 6, and 8 during Flat Run.

The data suggest that the WEC does not adversely impact grid stability during the flat run. While the WEC does not demonstrate a significant improvement in dynamic stability, it operates without introducing significant instability, highlighting its compatibility with the grid in steady-state conditions. In Figure 4 we observe the frequency of each bus following the wave height, however these oscillation amplitudes are all within an acceptable range (± 0.00035 p.u.).

B. Step Change Test

This section presents the results of the step-change test, where a sudden increase in active power (from 0.0 p.u. to 1.0 p.u.) was introduced at Bus 8 between 30 and 45 seconds. The comparison between the “with WEC” and “no WEC” scenarios highlights the impact of the WEC on grid stability during a disturbance.

1) *Active Power Response at Bus 8:* The primary impact of the step change event is observed at Bus 8, where the disturbance was introduced. In the “no WEC” case, the active power exhibits significant fluctuations during the step change period, as shown in Figure 5. In the “with WEC” scenario, the power response is marginally smoother. While this does not show a significant smoothing effect compared to the baseline, it indicates that the WEC can help buffer sudden demand shifts without destabilizing the system.

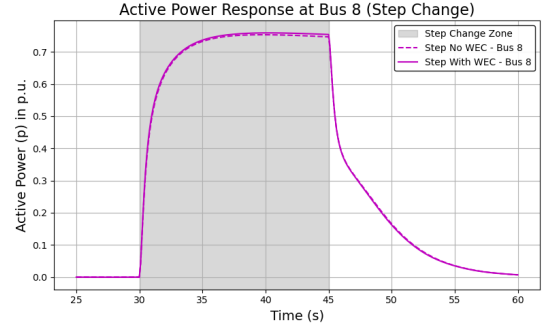


Fig. 5. Active Power at Bus 8 during Step Change.

2) *Frequency Response at Bus 8:* The frequency at Bus 8 (Figure 6) shows that the “no WEC” and “with WEC” scenario have essentially similar responses, with minimal noticeable difference.

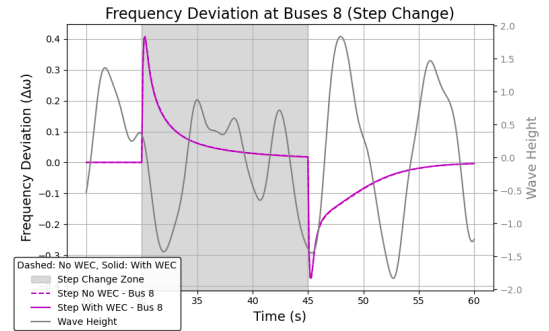


Fig. 6. Frequency at Bus 8 during Step Change.

3) *System-wide Generator Response: Active power in buses 1, 2, 3, and 6:* The step change event at Bus 8 also affects other generator buses (Buses 1, 2, 3, and 6). As seen in Figure 7, these buses experience slightly smoother transitions in active power when the WEC is integrated. The WEC’s influence is observed even in buses not directly involved in the disturbance, though the differences are minor. This suggests the WEC could potentially contribute to system-wide resilience, though further experiments are needed to measure any substantial stabilizing effect under a broad range of operating conditions.

4) *Frequency Response at Buses 1, 3, and 6:* In terms of frequency, the step change introduces noticeable deviations. However, the WEC supports the mitigation of these deviations, as seen in Figure 8. While the frequency response at Buses 1, 3, and 6 shows marginal improvement in the “with WEC”

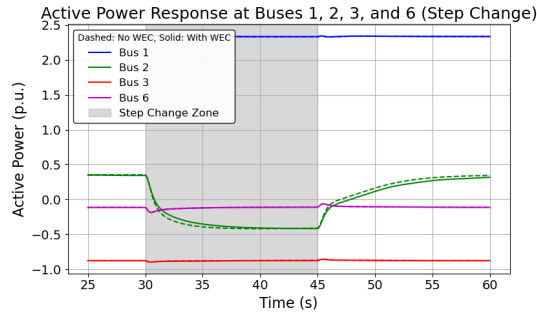


Fig. 7. Active Power at Buses 1, 2, 3, and 6 during Step Change.

scenario, the WEC's role in maintaining frequency stability during disturbances is limited. The results do not demonstrate significant smoothing, but the WEC does not introduce instability, aligning with a "no harm" result.

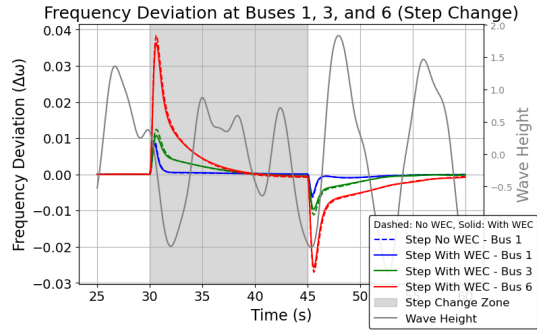


Fig. 8. Frequency at Buses 1, 3, and 6 during Step Change.

IV. CONCLUSION

In this work, we developed a co-simulation framework that supports the integration of WECs into grid operations such that dynamical analysis is rendered feasible. Our co-simulation framework is subsequently leveraged with practical scenarios to demonstrate how WECs can be effectively integrated into power grids without compromising stability. Specifically, the findings from the flat run test scenario example suggest that the WEC does not introduce substantial instability into the grid, with minimal differences in active power and frequency between the "with WEC" and "no WEC" scenarios. This establishes a solid baseline for WEC integration studies, working toward demonstrating that these devices can operate alongside traditional power systems without causing off-nominal fluctuations or disturbances. Under dynamic conditions, such as the step change event simulated in this study, the WEC showed marginal benefits. The results highlight the WEC's ability to potentially smooth out active power fluctuations and stabilize frequency, particularly at Bus 8, where the disturbance was introduced. Furthermore, the WEC's stabilizing effects extended across the grid, providing improved power flow and frequency stability at other generator buses (Buses 1, 3, and 6). The slack bus (Bus 2), responsible for maintaining system balance,

also benefited from WEC integration. The WEC helped reduce the magnitude of adjustments needed at the slack bus during dynamic events, contributing to a more efficient overall grid operation. These results reinforce the idea that WECs can play a vital role in future grid designs, particularly as the penetration of renewable energy increases.

In conclusion, as the demand for renewable energy continues to grow, WECs can become a critical resource to maintain reliable and resilient power systems, especially in regions with high variability for other sources of renewable generation. Future work should focus on further exploring the scalability of WEC integration and its impact on larger, more complex grid systems under a variety of operational conditions, as well as microgrids, for a different seasonal and resilience event scenarios.

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